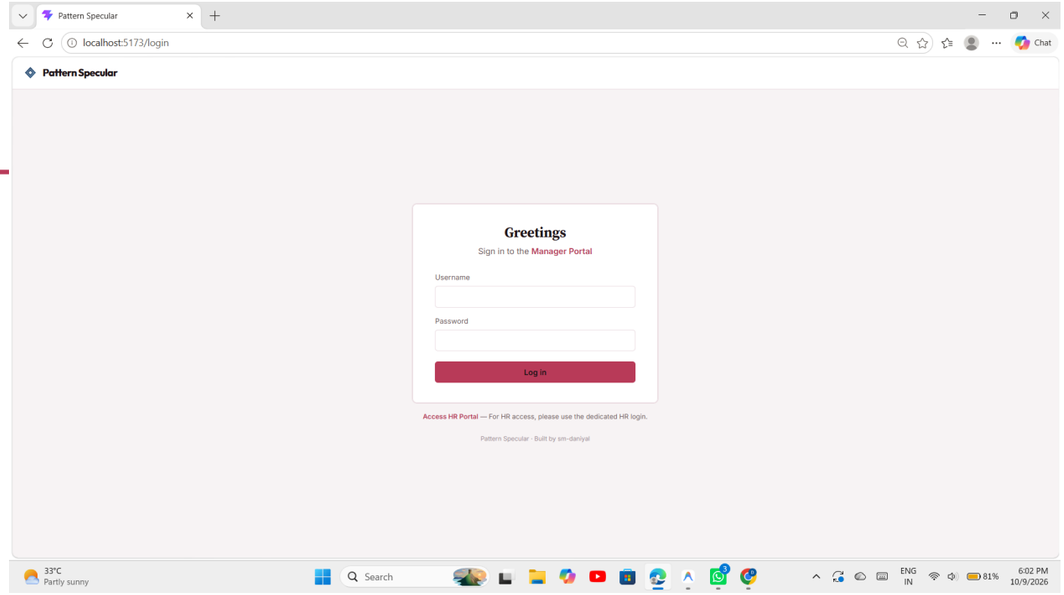
PATTERN SPECULAR

The Pattern Specular

“Surfaces bias patterns in a manager’s own writing, anchored in law and research, so they can decide what to change.”

Syed Daniyal (sm-daniyal) · Applied AI & Systems Engineering · India

Google Gemini 2.5 / 3.5 · React 19 · FastAPI · Neon PostgreSQL



ABOUT THE AUTHOR

Syed Daniyal (sm-daniyal)

What I build

- AI-native systems: LLM orchestration, multi-agent debate loops, schema-enforced structured outputs
- Modern full-stack: React 19, FastAPI async APIs, SSE streaming, TanStack Router / Query
- Statistical data science: exact finite-sample inference, differential privacy guarantees
- Multi-jurisdiction compliance: India, Singapore, US, UK labor law frameworks

github.com/sm-daniyal

Why I built Pattern Specular

Enterprises run annual unconscious-bias training, yet no tool assists the actual documents managers write — job descriptions, interview feedback, promotion cases. Bias in isolation looks like normal language. Patterns only emerge across months of writing.

Pattern Specular fills that gap: a sovereign, private, mathematically-grounded calibration mirror for each manager's own writing.

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THE PROBLEM

Where bias training stops, this begins.

The gap between understanding the principles and writing without bias

The training void

Managers complete annual fair-hiring modules, then return to writing job descriptions, interview feedback, and promotion cases with no live assistance. Training is theoretical; the writing is where decisions are codified.

Bias is invisible in one document

Describing one candidate as ‘supportive and polite’ and another as an ‘aggressive go-getter’ reads as perfectly normal in isolation. Patterns only emerge across dozens of evaluations over time.

The response

Pattern Specular flags statutory infractions in real time, checks feedback against the stated job criteria, and surfaces longitudinal tendencies only when Fisher’s Exact Test confirms significance ($p < 0.05$) — never as a punitive score.

DESIGN

Evidence in. Decision theirs.

A calibration mirror — never a gate, score, or judge

Every flag cites a statute or peer-reviewed paper. Every flag is dismissible. Submission is never blocked.

Capabilities

- Statutory flags: India Code on Wages 2019, RPwD Act 2016, TAFEP (SG), US Title VII, UK Equality Act 2010
- Criteria drift: checks feedback actually addresses the JD's stated criteria
- Longitudinal patterns: Fisher's Exact ($p < 0.05$) required before surfacing a habit
- Autonomous lexicon: new bias terms proposed by agents, approved by HR humans

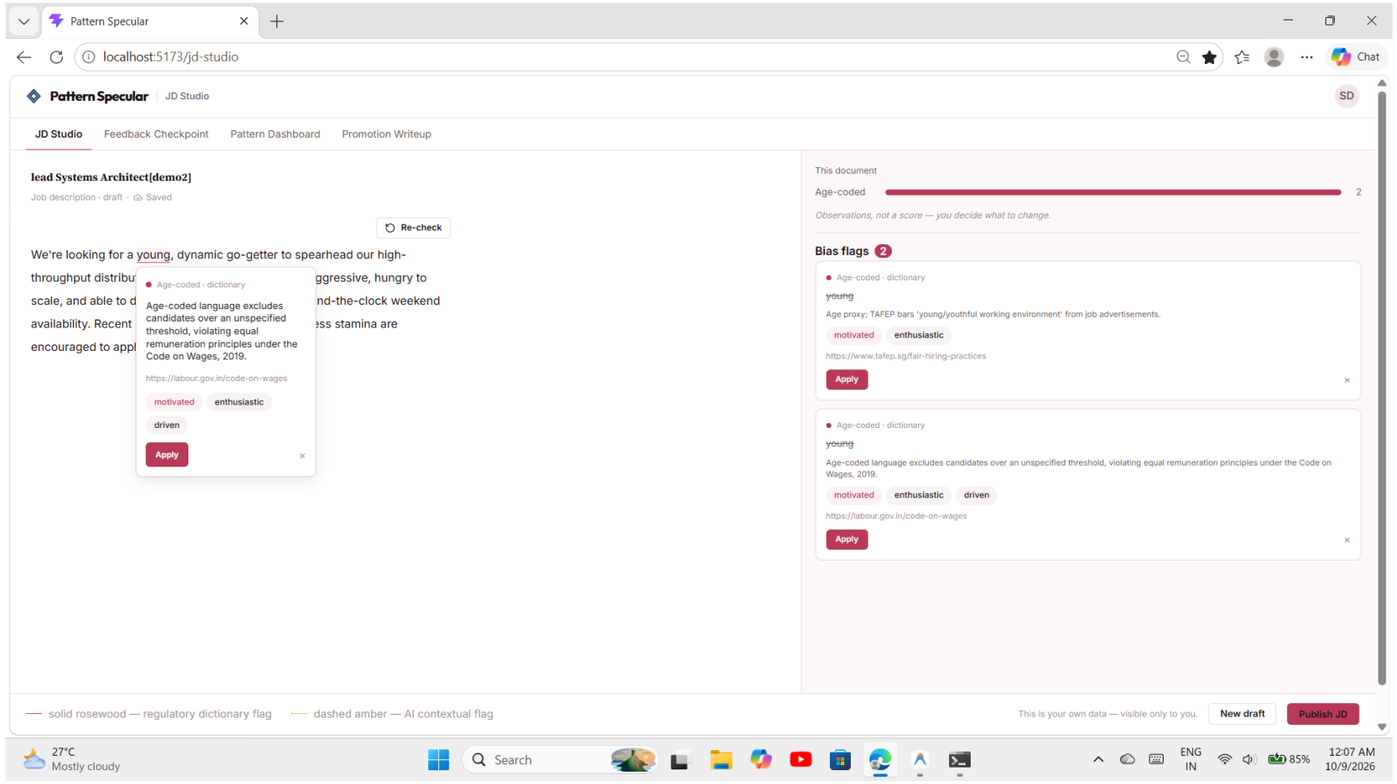
Non-negotiable invariants

- Mirror, not judge — no scores, grades, or penalties
- Private by architecture — HR cannot read any manager's writing
- Non-blocking — managers dismiss freely and retain full authority
- Not a chatbot, not a training module, not a hiring gate

PRODUCT · MANAGER

Three writing surfaces, one detection pipeline.

Sub-10ms deterministic lemmatizer + async Gemini contextual pass



JD Studio

Live statutory flags as you type. Rosewood citation cards with rewrite chips.

Feedback Checkpoint

Checks interview notes for gendered language and criteria drift against the original JD.

Promotion Writeup

Validates promotion cases against rubric criteria and peer feedback, surfacing under-evidenced claims.

How detection works

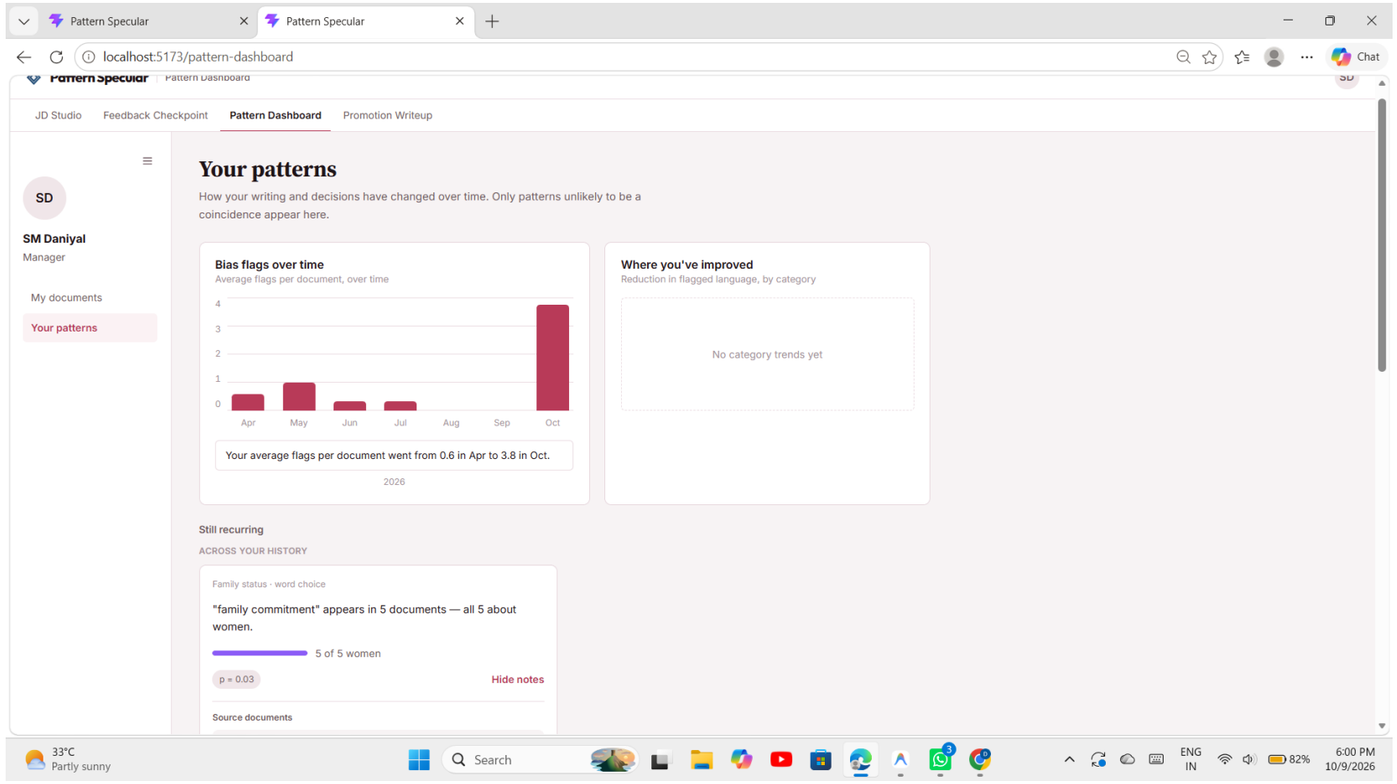
— Stage 1 (<10ms): spaCy lemmatizer matches statutory terms exactly — zero LLM cost

— Stage 2 (async): Gemini evaluates coded language and genuine occupational exceptions in context

PRODUCT · MANAGER

A pattern is a proof, not a count.

Fisher’s Exact Test gates every finding before it surfaces



Longitudinal trend view

12-month bar chart across age, gender, disability, family status categories.

Fisher’s Exact Test gating

‘Family commitment appears in 5 documents — all 5 about women’ triggers only when the 2×2 contingency table yields $p = 0.03 < 0.05$. Five flags in 100 documents is ignored as noise.

Behavioral adoption tracking

Charts what fraction of flags were voluntarily revised vs. dismissed over time.

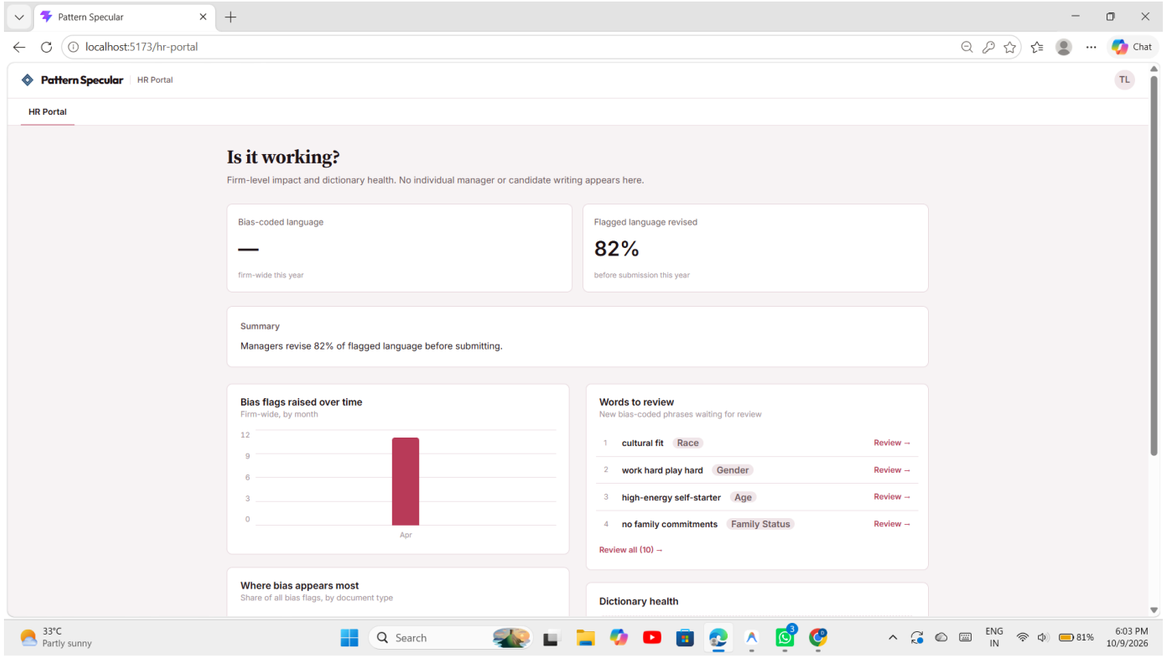
Private to the manager

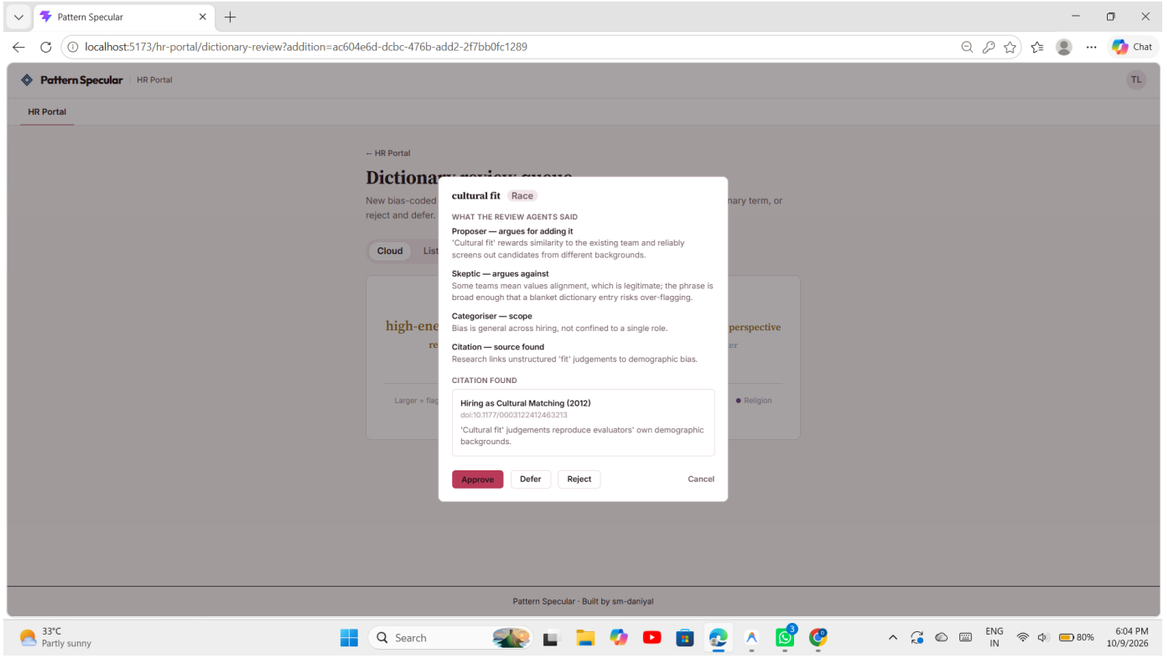
Backed by strict owner_id database tenancy — not accessible to HR.

PRODUCT · HR

HR sees firm trends. Not manager writing.

k ≥ 3 cell suppression — privacy enforced at the query layer, not by policy





Left: Firm-wide adoption portal — 82% revision rate, monthly volume, document type mix. Any filter with <3 distinct managers is suppressed (k ≥ 3 anonymity in SQL HAVING clause). Right: Dictionary review queue — 4-agent debate transcript and HR approve/reject button.

7 stages. Every flag must survive each one.

Deterministic guardrails prevent hallucinated flags from reaching users

1 · Dictionary — spaCy lemmatizer (<10ms). Exact match against citation-backed statutory lexicons. Zero LLM cost.

2 · Contextual — Gemini 3.5 Flash Lite call. Catches coded, non-literal language; evaluates GDOR exceptions.

3 · Adjudicator — Verbatim span check in Python. Any quote not found exactly in the source text is dropped — hallucinations cannot pass.

4 · Verdict — Statutory rulings applied in context. Confirmed bona-fide occupational requirements suppressed.

5 · Suppression — SHA-256 sentence fingerprints checked against the manager's historical dismissals. Previously dismissed, unchanged context → suppressed.

6 · Judge — Multi-sample Gemini consensus. Flags with agreement rate <70% are terminated here.

7 · Recommendations — 2–3 evidence-grounded neutral rewrite chips per confirmed flag.

ENGINEERING · JUDGE

A second model must independently agree.

Adversarial confidence gate eliminating generator self-preference bias

Independent model

The Judge evaluates flags without seeing the generator's explanation, eliminating LLM self-preference bias.

Empirical confidence

Confidence = agreement fraction across N independent samples. Flags with <70% agreement are discarded.

Rubric decomposition

Not a 0–5 score. Boolean criteria: references_characteristic, gdor_plausible, stated_objectively.

Position shuffling

Flag order randomised before grading to eliminate list-position primacy bias.

Adjudication formula: biased ☐ references_characteristic ☐ ¬(gdor_plausible ☐ stated_objectively)

ENGINEERING · RESEARCH

Five published papers, one design.

Every architectural choice is grounded in empirical ML research

Panickssery, Bowman & Feng · 2024 · arXiv:2404.13076

Self-preference bias in LLM judges is causal. The Judge stage uses a different model from the generator.

CheckEval — Lee et al. · EMNLP 2025 · arXiv:2403.18771

Boolean checklist rubrics have lower variance than holistic 1–5 scoring. Adopted as the Judge's rubric decomposition.

Wang et al. · ICLR 2023 · arXiv:2203.11171

Self-consistency via majority voting over multiple samples matches logprob quality. Basis for frequency-based confidence.

Zheng et al. · NeurIPS 2023 · arXiv:2306.05685

LLM judges favour items by list position. Flag order is randomised before every Judge prompt.

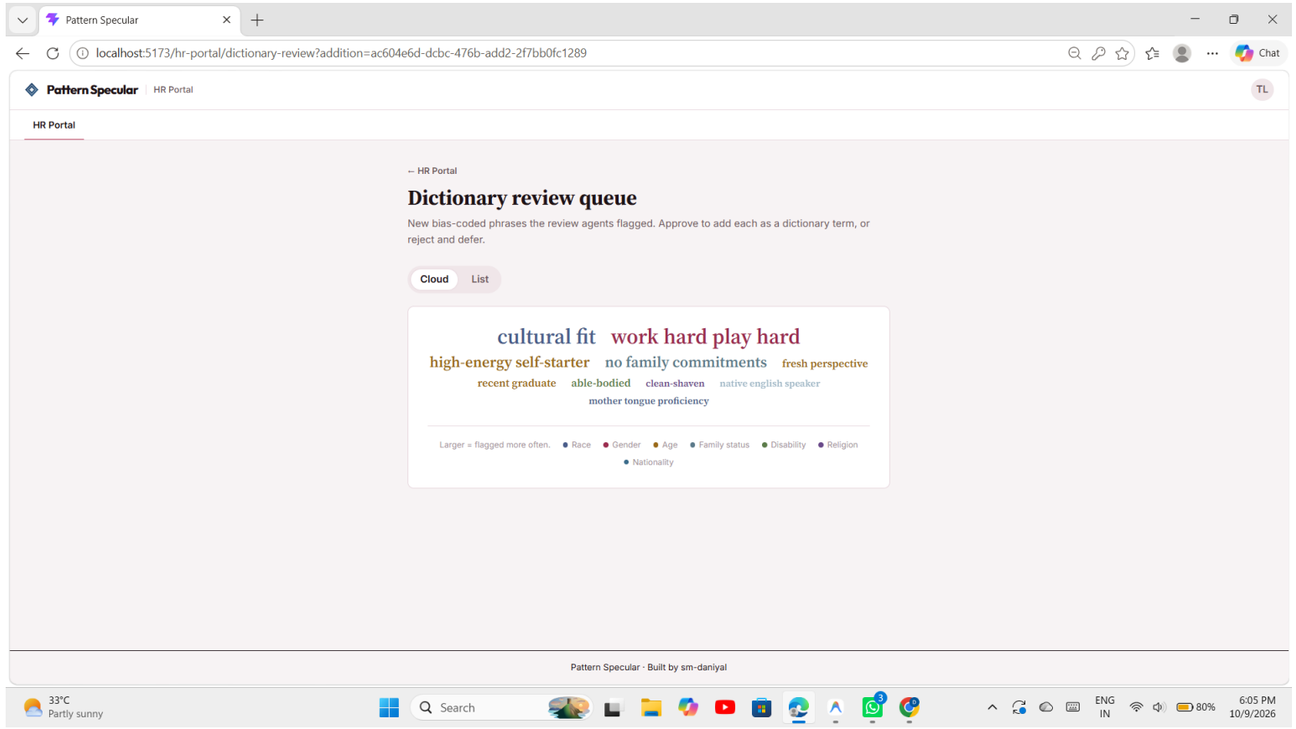
Tian et al. · EMNLP 2023 · arXiv:2603.05485

Calibration curves (ECE, Brier score) against gold labels set the 0.70 confidence threshold.

ENGINEERING · DICTIONARY

The lexicon grows itself.

4-agent debate → HR approval → live dictionary entry



The 4-agent committee

- Proposer: argues the phrase introduces protected-class exclusion
- Skeptic: tests whether the term has valid standard-business meaning
- Categoriser: scopes general vs. role-specific applicability
- Citation agent: finds a statute or peer-reviewed source — no fabrication

Gate conditions (all three required)

- Citation verified
- Scope is general — not just one role type
- Proposer or Skeptic verdict: in favour

Human sign-off

HR reviews the full debate transcript and clicks Approve or Reject. Approved terms enter the live dictionary, making future hits sub-10ms Stage 1 matches with zero LLM cost.

ENGINEERING · ARCHITECTURE

One API, one SPA, one serverless database.

Decoupled cloud-native layers — each independently scalable

Frontend

- — React 19 + Vite — concurrent rendering, instant HMR
- — TipTap / ProseMirror — headless rich text with custom flag marks
- — Tailwind CSS v4 — Rosewood atomic token system
- — TanStack Router + Query — type-safe routes and server-state cache

Backend

- — Python 3.12 + FastAPI — async REST + SSE streaming
- — LangGraph StateGraph — deterministic 7-stage machine
- — spaCy en_core_web_sm — in-memory tokenization (<10ms)
- — scipy.stats — Fisher's Exact hypergeometric solver

Data & AI

- — Neon Serverless PostgreSQL 16 — auto-scaling, connection pooling
- — SQLAlchemy 2.0 + Alembic — typed ORM, 20 migrations
- — Google Gemini 3.5 Flash Lite — structured output via Instructor
- — Pydantic v2 — schema-enforced AI boundaries

ENGINEERING · DATA MODEL

Privacy is a schema constraint, not a policy.

17 relational entities encoding structural guarantees

Core entities

User & UserRoleAssignment · Document & AnalysisRun · Flag, FlagDismissal, FlagInteraction · Dictionary & Citation (FK non-nullable) · DriftFinding, PeerFeedback, PromotionRubricCriterion · AgentRun (full LLM cost audit trail)

Three structural privacy boundaries

— owner_id: every manager read is parameterised by owner. No endpoint exists to query another manager's documents.

— Mandatory Citation FK: a dictionary entry with no statutory evidence is a foreign-key violation — impossible to create.

— AgentRun audit: every LLM prompt, token count, and latency is a relational fact, not a log line.

Version 2 seams are already built in.

Extensibility without touching the engine core

Multi-model gateway

StructuredCompletionClient protocol already abstracts LLM providers. V2 enables runtime routing across Gemini, Claude, and local models.

RAG over historical writing

pgvector semantic search across past evaluations, scoped to owner_id. Documents are stored; adding the index is a migration.

Multimodal video drift

Ingest video interview recordings into Gemini’s native AV context window. DriftFinding.reference_kind already has the enum.

Global statutory plug-ins

EU AI Act, Australia Fair Work Act: new region-scoped citation rows only. No engine code changes.

What building this taught me.

Spec-driven development pays off

Writing the behavioural specification before code kept scope sharp, made implementation faster, and gave reviewers a single source of truth.

Statistical rigour prevents fatigue

Requiring Fisher's Exact Test significance before surfacing a pattern means managers trust what they see rather than tuning it out.

Schema-first AI boundaries

Pydantic v2 + Instructor enforcing structured outputs across LLM calls eliminated an entire class of JSON malformation bugs.

Treat prompts as a testable code surface

Prompt changes go through the same review, diff, and regression-test discipline as source code.

Privacy by construction, not policy

Moving privacy invariants into the relational schema (owner_id, non-nullable FK) makes violations structurally impossible, not just forbidden.

Human-in-the-loop is a feature, not a fallback

The HR dictionary approval queue makes the system safer and more trusted than a fully autonomous one.

THANK YOU

See the code. Get in touch.

Pattern Specular is an independent applied-AI portfolio project

Repository

github.com/sm-daniyal/patternSpecular

GitHub profile

github.com/sm-daniyal

What's in the repo

- docs/DESIGN_SPEC.md — full 14-section product specification
- docs/architecture/ — flow, sequence, and ER diagrams
- docs/adr/ — 18 architecture decision records
- docs/references/compliance_frameworks.md — statutory grounding (IN, SG, US, UK)
- docs/demo/DEMO_FLOW.md — 10-minute live demo script

Designed, implemented, verified, and calibrated by Syed Daniyal.